endoscopist and CADx agreed on a diagnosis were retained. Primary outcome was sensitivity for adenoma diagnosis of CADx assisted optical diagnosis based on the agreement framework versus CADx unassisted optical diagnosis, using pathology results as a reference standard. Secondary outcomes included accuracy, diagnostic characteristics, and optical diagnosis surveillance interval agreement with pathology-based United States Multi-Society Task Force (USMSTF) guidelines. Results: A total of 810 polyps, of which 444 and 366 underwent CADx assisted and CADx unassisted optical diagnosis respectively, were included in our analysis. In CADx assisted cases, endoscopists and CADx agreed in their diagnosis in 72.3% of cases, meaning 321/444 diminutive polyps could undergo optical diagnosis according to an agreement-based approach. Sensitivity for adenoma diagnosis in these polyps (adenomas, n=234) was 93.4% (95% CI 89.9-96.8) versus 83.8% (95% CI 77.9-89.7) for the 366 polyps (adenomas, n=245) undergoing CADx unassisted optical diagnosis (p=0.008). Diagnostic accuracy of agreement-based CADx assisted and CADx unassisted optical diagnosis was 82.2% (95% CI 78.0-86.3) and 74.8% (95% CI 69.5-80.0) respectively (p=0.026). Diagnostic characteristics are summarized in Table 1. Excluding cases with CADx and endoscopist disagreement from CADx assisted optical diagnosis filtered out cases with low diagnostic accuracy (123 polyps; accuracy 35.0% (95% CI 26.2- 43.8)). Conclusions: The CADx and endoscopist agreement framework effectively filters out cases with low diagnostic accuracy from undergoing CADx assisted optical diagnosis. Applying this framework allows endoscopists and CADx to work in synergy resulting in diagnostic performance superior to CADx unassisted optical diagnosis performance.

Figure 1. CADx assisted optical diagnosis implementation frameworks

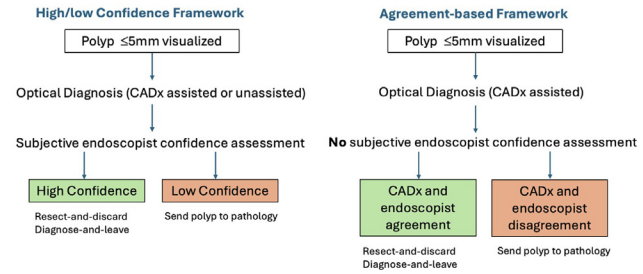


Table 1. Diagnostic performance of optical diagnosis strategies

	CADx assisted (agreement-based)	CADx unassisted	p values
Polyps, n	321	366	
Accuracy, % (95% CI)	82.2 (78.0-86.3)	74.8 (69.5-80.0)	**0.026**
Diagnostic characteristics, % (95% CI)			
Sensitivity	93.4 (89.9-96.8)	83.8 (77.9-89.7)	**0.008**
Specificity	63.3 (52.1-74.4)	57.7 (48.2-67.3)	0.41
PPV	85.8 (81.2-90.5)	82.9 (77.5-88.4)	0.41
NPV	70.5 (59.6-81.4)	70.5 (59.6-81.4)	0.2

CONTROL ID: 4235745

A NOVEL WEB APPLICATION UTILIZING GENERATIVE ARTIFICIAL INTELLIGENCE TO ENHANCE ENDOSCOPY EDUCATION FOR GASTROENTEROLOGY FELLOWS

Shaleen Vasavada, Theresa H. Nguyen, Scott Larson

ASGE Oral/Poster: Disclosure Compliance: I understand. DDW 2025 PARTICIPANT DISCLOSURE FORM: Shaleen Vasavada: NO financial relationship with a commercial interest | Theresa Nguyen: NO financial relationship with a commercial interest | Scott Larson: No Answer.

Abstract Body: Background: Competency in endoscopic procedures is essential for gastroenterology fellows, with tools like the American Society for Gastrointestinal Endoscopy's (ASGE) Assessment of Competency in Endoscopy (ACE) providing critical metrics for evaluating performance. We developed a mobile-friendly evaluation platform incorporating ACE tools, enhanced with Large Language Model (LLM)-based feedback. This study investigates the impact of LLM-generated personalized feedback on fellow endoscopy performance. Methods: A cohort of gastroenterology fellows (N=24) was divided into junior and senior fellows, with fellows randomly assigned to intervention (LLM feedback) or control (no feedback) groups. The platform utilized ACE tools for both esophagogastroduodenoscopy (EGD) and colonoscopy assessments, adapted to deliver individualized feedback derived from a curated knowledge base of endoscopy textbooks and peer-reviewed articles (Figure 1). Key endpoints included improvements in cognitive and motor skills, mucosal visualization, and tip control over a 6-month period. Statistical analyses included t-scores and linear regression to evaluate group differences and trends. Results: The analysis revealed that the intervention group, which received LLM-based feedback, showed significant improvements in several key areas. For junior fellows, there was a statistically significant increase in aggregate motor skill scores in the intervention group compared to the control group ($p = 0.032$) (figure 2). Cognitive skills also improved significantly among junior fellows ($p = 0.003$), while mucosal visualization and tip control trended towards improvement with LLM-generated feedback. Senior fellows, however, showed minimal differences between intervention and control groups, with no statistically significant difference observed for motor skills ($p = 0.895$) or mucosal visualization ($p = 0.808$). Regression analysis demonstrated a significant effect of seniority on skill performance, with strong trends across all skill domains ($p < 0.001$ for cognition, motor skills, mucosal visualization, and tip control). Conclusions: LLM-based personalized feedback enhances performance in endoscopic evaluations, particularly among junior fellows, where the most substantial improvements were observed in cognitive and motor domains. This novel, scalable approach highlights the potential of AI-assisted education to optimize training outcomes in gastroenterology.

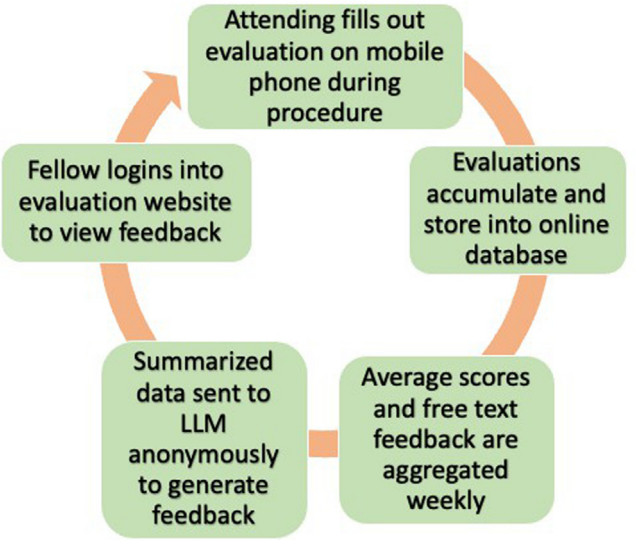


Figure 1.

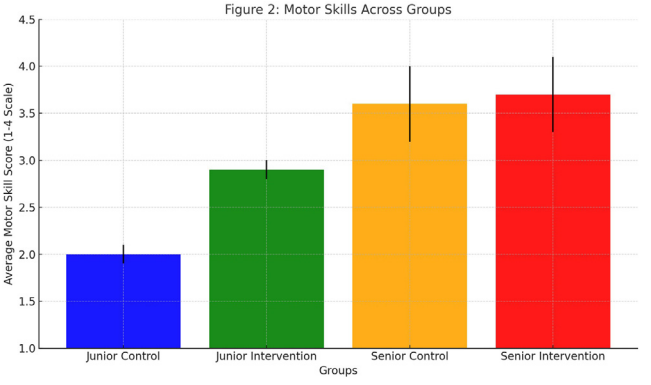


Figure 2.

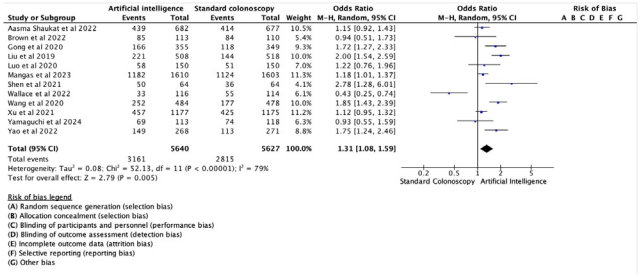
CONTROL ID: 4255343

REAL-TIME ARTIFICIAL INTELLIGENCE VS STANDARD COLONOSCOPY IN THE EARLY DETECTION OF COLORECTAL CANCER: A META-ANALYSIS

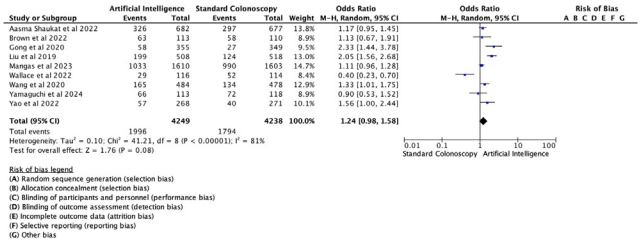
Rahul Chikatimalla, Sunny Kumar, Binay K. Panjiyar, Suchith B. Suresh, Osama A. Rehman, Aasta kumari, Rahul Akkapeddi

ASGE Oral/Poster: Disclosure Compliance: I understand. DDW 2025 PARTICIPANT DISCLOSURE FORM: Rahul Chikatimalla: NO financial relationship with a commercial interest | Sunny Kumar: NO financial relationship with a commercial interest | Binay Panjiyar: NO financial relationship with a commercial interest | Suchith Suresh: NO financial relationship with a commercial interest | Osama Rehman: NO financial relationship with a commercial interest | Aasta kumari: NO financial relationship with a commercial interest | Rahul Akkapeddi: NO financial relationship with a commercial interest

Abstract Body: Background & Objectives: Colonoscopy remains the gold standard for colorectal cancer prevention by identifying and removing precursor lesions. Deep learning systems with real-time computer-aided polyp detection (CADe) have demonstrated high accuracy in artificial settings, and preliminary randomized controlled trials (RCTs) have reported favorable outcomes in clinical settings. However, inconsistent findings across RCTs necessitate a comprehensive analysis. This meta-analysis aims to evaluate the efficacy of AI-assisted colonoscopy compared to standard techniques, focusing on Polyp Detection Rate (PDR) and Adenoma Detection Rate (ADR), and to explore their implications for clinical practice. Methods: A systematic search was conducted in Google Scholar, EMBASE, Cochrane Central, MEDLINE/PubMed, ClinicalTrials.gov, and CINAHL (2019–November 2024) for RCTs comparing AI-assisted to standard colonoscopy. Random-effect models were utilized to calculate pooled odds ratios (ORs) with 95% confidence intervals. The risk of bias was assessed using the Cochrane Risk of Bias Tool, and heterogeneity was quantified using I^2 statistics. Results: From 22,762 studies, 12 RCTs (n = 11,267) met the inclusion criteria. AI-assisted colonoscopy significantly improved PDR with an odds ratio (OR) of 1.31 (95% CI, 1.08–1.59; p = 0.005), despite heterogeneity among studies (I^2 = 79%). While ADR showed improvement with AI-assisted colonoscopy (OR 1.24; 95% CI, 0.98–1.58), the result was not statistically significant (p = 0.08) and had high heterogeneity (I^2 = 81%). The age and gender of participants were similar in both the AI-assisted and control groups across all the studies. The mean age ranged from 49 to 63 years across studies, and no significant differences were noted between AI and control groups. The results suggest that AI- assisted colonoscopy offers improved polyp detection, while the benefits for adenoma detection require further investigation. Conclusion: AI-assisted colonoscopy significantly enhances PDR, highlighting its potential role in colorectal cancer screening programs. However, while an improvement in the ADR was observed, the results were not statistically significant and showed considerable variability. These findings highlight the promise of AI in improving diagnostic accuracy but also point to the need for further research to better understand its impact on meaningful clinical outcomes. Incorporating AI into routine colonoscopy could play a key role in advancing colorectal cancer screening and prevention efforts.



Forest plot of comparison: Artificial Intelligence vs. Standard colonoscopy in Early Detection of Colorectal Cancer, outcome: Polyp detection rate.



Forest plot of comparison: Artificial Intelligence vs Standard Colonoscopy in early detection of colorectal cancer, outcome: Adenoma detection rate.

CONTROL ID: 4225216

ARTIFICIAL INTELLIGENCE-BASED MARKS DETECTION AND INCISION GUIDE LINE PREDICTION MODEL IN ESOPHAGEAL ENDOSCOPIC SUBMUCOSAL DISSECTION: A MULTICENTER STUDY

Ruide Liu, Xianglei Yuan, Kaide Huang, Zhang Yi, Bing Hu

ASGE Oral/Poster: Disclosure Compliance: I understand. DDW 2025 PARTICIPANT DISCLOSURE FORM: Ruide Liu: NO financial relationship with a commercial interest | Xianglei Yuan: No Answer. | Kaide Huang: No Answer. | Zhang Yi: No Answer. | Bing Hu: No Answer.

Abstract Body: Background: Endoscopic submucosal dissection (ESD) was an important minimally invasive procedure for treating early esophageal cancer, where surrounding mucosal incision (SMI) was a crucial yet challenging phase. Misidentification or omission of the marks may result in incorrect incision, which may lead to serious consequences such as incomplete dissection and postoperative esophageal stenosis. Therefore, we aimed to developed an artificial intelligence (AI) model for marks detection and incision guide line prediction during SMI, and to evaluate its performance compared to multicenter endoscopists. Methods: Images were extracted at one frame per second from SMI video clips and divided into 3 datasets: training dataset, internal and external test dataset for cross-validation and testing of the AI model. Twenty-two endoscopists from 5 hospitals were invited to participate in the comparison with model, including eight senior and fourteen junior endoscopists. The precision, recall, F1 score, accuracy, average distance error (ADE), and a marks-guide line subjective score (M-GSS) were used for outcomes. Results: A total of 46280 SMI images were extracted from 166 esophageal ESD videos in 4 hospitals between 2016 and 2024. For marks detection, the precision of the AI model in the training dataset, internal, and external test dataset were 78.97%, 85.76%, and 79.46%, respectively. For the incision guide line prediction, the ADE was 0.096, 0.097, and 0.159, respectively. The AI model achieved an accuracy (63.21% vs 54.57%, P<0.001), ADE (0.081 vs 0.241, P=0.002), and M-GSS (1.88 vs 1.75, P<0.001) significantly better than that of junior endoscopists, and were comparable with those of senior endoscopists. Conclusions: The AI model achieved promising results on large SMI image datasets, showing comparable effectiveness with senior endoscopists for marks detection and incision guide line prediction, which had the potential to navigate SMI safely and properly.

Table 1 Performance of the AI model on datasets

		Training dataset	Internal test dataset	External test dataset
Marks	Precision, %	78.97 (75.76-82.19)	85.76	79.46
	Recall, %	68.01 (64.27-71.74)	69.63	55.87
	F1 score, %	73.01 (71.55-74.46)	76.86	65.61
	Accuracy, %	57.50 (55.70-59.30)	62.41	48.82
Incision guide line	ADE	0.096 [0.007-1.000]	0.097 [0.007-1.000]	0.159 [0.014-1.000]
Negative image	TNR, %	81.22 (78.09-84.35)	86.30	83.44

Number or mean (95% confidence interval) or median [interquartile range]

ADE, average distance error; TNR, true negative rate.

Figure 1. Table: Performance of the AI model on datasets